

Notes: Morellec, Nikolov, and Schürhoff (JF, 2012)

Corporate Governance and Capital Structure Dynamics

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
2.1	What was known before this paper	2
2.2	What this paper adds	3
2.3	Why the contribution matters	3
3	Model objects, data, and measurement	3
3.1	Firm cash flows and the state variable	3
3.2	Debt, taxes, and restructuring	3
3.3	Default, liquidation, and renegotiation	4
3.4	Managerial ownership and private benefits	4
3.5	Empirical sample and variables	4
4	Models and theoretical specifications	4
4.1	Managerial objective	4
4.2	Capital-structure regions	5
4.3	Model without agency conflicts	5
4.4	Model with agency conflicts	5
4.5	Refinancing-cost benchmark	5
4.6	Governance regressions	5
5	Identification and empirical design	6
5.1	What identifies agency costs in the structural model	6
5.2	Why refinancing costs and agency costs are separable	6
5.3	Why governance regressions matter	6
5.4	Limitations of the design	6
6	Section-by-section study guide	7
7	Tables and figures	7
7.1	Table I: Comparative Statics + Table II: Calibrated Leverage Distribution and Model Identification	7
7.2	Figure 1: Monte Carlo Simulation Precision and Accuracy	8
7.3	Table III: Variable Definitions + Table IV: Descriptive Statistics	10
7.4	Table V: Refinancing Cost Estimates Without Agency Conflicts + Table VI: Structural Parameter Estimates With Agency Conflicts	11
7.5	Figure 2: Predicted Agency Costs Across Firms + Figure 3: Cost of Debt and Underleverage	11

7.6	Table VII: Distribution of Debt Costs Across Firms + Table VIII: Robustness of Agency Cost Estimates	12
7.7	Table IX: Conditional Moment Tests + Table X: Simulation Evidence on Leverage Inertia and Mean Reversion	13
7.8	Table XI: The Determinants of Agency Conflicts	14
8	Detailed result map	15
9	Short summary	16
10	Conclusion	16
10.1	Core conclusion	16
10.2	Economic intuition	16
10.3	Methodological contribution	16
10.4	Relation to the capital-structure literature	16
10.5	Corporate governance implications	17
10.6	Limitations	17
10.7	Takeaway	17

1 Big picture

- **Question.** How important are manager–shareholder conflicts for capital structure choices and capital structure dynamics?
- **Core puzzle.** Standard tax-benefit models predict more leverage than many firms actually use. Dynamic tradeoff models with refinancing costs can generate inertia, but refinancing costs alone have difficulty matching the low average leverage, the persistence of leverage, and the cross-sectional dispersion in leverage.
- **Main mechanism.** Managers can divert a fraction of free cash flow to equity as private benefits. Debt disciplines managers by reducing free cash flow and limiting diversion. But because the manager personally dislikes the disciplining effect of debt, the manager chooses lower leverage than shareholders would choose.
- **Model environment.** Firms face corporate and personal taxes, refinancing costs, liquidation costs, renegotiation costs, and dynamic default/restructuring decisions. Cash flows follow a geometric Brownian motion. Debt is perpetual and callable, and the manager controls financing, restructuring, and default policy.
- **Key structural innovation.** The paper estimates firm-specific agency conflicts from leverage dynamics, not just from leverage levels. It uses the mean, dispersion, persistence, tails, and other moments of leverage to identify refinancing costs and agency costs.
- **Main estimate.** The average cost of managerial discretion is about 1.5% of equity value, with a median of about 0.45%. These costs are small enough to be plausible but large enough to resolve the low-leverage puzzle.
- **Debt issuance cost result.** A model without agency conflicts would require very large refinancing costs, roughly 12.6% of total debt outstanding, or around 25% of the amount issued, to explain observed leverage dynamics. This is much larger than typical empirical issuance-cost estimates.
- **Governance result.** The structurally estimated private benefits of control are lower when governance is stronger: institutional ownership, independent blockholders, and board independence are associated with lower agency conflicts. CEO tenure and antitakeover provisions are associated with higher agency conflicts.
- **Incremental contribution.** The paper takes a mostly qualitative agency-cost literature and turns it into a quantitative, structurally estimated explanation of observed leverage levels and leverage dynamics.
- **Takeaway.** Capital structure is not just a tradeoff between taxes and bankruptcy costs. It is also a dynamic governance problem: leverage is partly chosen by managers who internalize private benefits and private costs differently from shareholders.

2 Incremental contribution of the paper

2.1 What was known before this paper

Prior research had established several pieces of the capital-structure puzzle:

- Static and dynamic tradeoff models explain why debt creates tax benefits and distress costs.
- Dynamic tradeoff models explain why firms do not continuously adjust leverage: adjustment costs and refinancing frictions create leverage inertia.
- Agency theories, especially Jensen (1986), Stulz (1990), and Zwiebel (1996), argue that managers may choose suboptimal leverage because debt limits discretion.
- Empirical work showed that leverage is persistent and that standard observables explain only part of cross-sectional capital-structure variation.

2.2 What this paper adds

- **Dynamic agency-capital-structure model.** The paper embeds managerial private benefits into a dynamic tradeoff model with debt issuance, restructuring, default, liquidation, and renegotiation.
- **Quantification of agency conflicts.** Instead of only predicting the direction of agency effects, it estimates their economic magnitude from the data.
- **Moment-based structural identification.** The paper uses many moments of leverage – levels, dispersion, persistence, skewness, tails, changes, and event frequencies – to identify the model. This is richer than a regression of leverage on governance variables.
- **Agency-cost interpretation of persistent leverage heterogeneity.** It gives an economic interpretation to persistent firm-specific leverage differences: part of the heterogeneity reflects persistent differences in manager–shareholder conflicts.
- **Governance validation.** It validates the estimated agency-cost measure by showing that it correlates in intuitive ways with governance proxies such as institutional ownership, independent blockholders, CEO tenure, antitakeover provisions, and board independence.
- **Distinguishing adjustment costs from agency costs.** It shows that refinancing costs alone can fit some inertia patterns only at implausibly high values, whereas modest agency costs can jointly match leverage levels and dynamics.
- **Manager-versus-shareholder cost distinction.** The model separates the cost of debt to shareholders from the cost of debt as perceived by managers. This is crucial because the decision maker is not the same as the residual claimant.

2.3 Why the contribution matters

The paper changes the empirical capital-structure question from “Do governance variables correlate with leverage?” to “How large must agency conflicts be to rationalize the entire stochastic process of leverage?” This is a stronger test because the same structural parameters must explain many leverage moments at once. It also shows why dynamic adjustment frictions and governance frictions should be studied jointly: adjustment costs create inertia around a target, while agency conflicts move the target itself and affect the distribution of leverage around it.

3 Model objects, data, and measurement

3.1 Firm cash flows and the state variable

The firm-specific operating cash flow is denoted X_{it} and follows a geometric Brownian motion under the risk-neutral measure:

$$dX_{it} = \mu_i X_{it} dt + \sigma_i X_{it} dB_{it}. \quad (1)$$

Detailed interpretation: Cash flow is the fundamental state variable. It determines the ability to service debt, the value of tax shields, the likelihood of default, and the option value of waiting before refinancing. Because cash flow evolves stochastically, leverage is a stochastic process, not a fixed target number.

3.2 Debt, taxes, and restructuring

Debt is modeled as perpetual debt with coupon c and principal P . Debt is callable and issued at par. Firms can increase debt by refinancing and paying a proportional cost λ , but they can reduce debt only in default. Cash flows are taxed at the corporate rate τ_c , while coupon and equity cash flows face personal taxes τ_d and τ_e .

Detailed interpretation: This structure creates a standard dynamic tradeoff: debt provides tax benefits, but it also raises default risk and refinancing costs. The key extension is that leverage also reduces managerial discretion.

3.3 Default, liquidation, and renegotiation

When cash flows fall sufficiently, debt is in distress. Default leads either to liquidation, which destroys fraction α of asset value, or renegotiation, which destroys fraction $\kappa < \alpha$. Shareholders and creditors bargain over the renegotiation surplus, with shareholder bargaining power denoted η .

Detailed interpretation: Liquidation and renegotiation matter because they determine the deadweight cost of debt and the value of default options. They also influence the manager's willingness to take on debt, because default disciplines or removes control.

3.4 Managerial ownership and private benefits

Managers own a fraction φ of equity and can divert a fraction ϕ of free cash flow to equity as private benefits. The parameter ϕ is the core agency-conflict parameter estimated by the model.

Detailed interpretation: Higher φ aligns managers with shareholders because the manager bears more of the wealth consequences of debt policy. Higher ϕ increases the value of control and makes managers more reluctant to issue debt.

3.5 Empirical sample and variables

The empirical sample contains 13,159 firm-quarter observations for 809 nonfinancial, nonregulated firms from 1992 to 2004. The authors combine Compustat, CRSP, I/B/E/S, ExecuComp, IRRC governance, IRRC blockholder/director data, and Thomson Financial institutional ownership data.

Detailed interpretation: The data are designed to support two separate tasks: first, estimate the structural model from leverage dynamics; second, compare the inferred agency-cost estimates to observed corporate-governance mechanisms.

4 Models and theoretical specifications

4.1 Managerial objective

The manager chooses financing, restructuring, and default policies. Because the manager owns only a fraction φ of equity and captures private benefits ϕ , the manager maximizes a payoff that differs from shareholder value.

$$\text{Manager payoff} = \varphi \times \text{equity value} + \text{private benefits of control.} \quad (2)$$

Detailed interpretation: Debt has two effects on the manager. It gives tax benefits and can increase shareholder value, but it also reduces free cash flow and therefore reduces the manager's ability to capture private benefits. This makes the manager's optimal leverage lower than the shareholder-optimal leverage.

4.2 Capital-structure regions

The dynamic model has three key boundaries:

- a default boundary, where the firm enters default or renegotiation;
- a restructuring/refinancing boundary, where the firm calls debt and issues more debt;
- a target leverage level selected after refinancing.

Detailed interpretation: The key implication is that leverage has a distribution. Firms drift away from target leverage as cash flows move. They refinance only when the value gain from moving back to target exceeds the refinancing cost.

4.3 Model without agency conflicts

When $\phi = 0$, the manager has no private benefit of control. Financing decisions are governed by taxes, liquidation costs, renegotiation costs, and refinancing costs.

Detailed interpretation: In this benchmark, matching the low observed leverage ratios requires implausibly high debt issuance costs. Thus, adjustment costs alone are not enough quantitatively.

4.4 Model with agency conflicts

The structural agency-cost parameter is estimated using

$$\phi_i = h(\alpha_\phi + \varepsilon_i^\phi), \quad (3)$$

where h maps the latent index into the admissible range and ε_i^ϕ is a firm-specific unobserved determinant of private benefits.

Detailed interpretation: The model estimates firm-specific private benefits of control from leverage behavior. A firm with persistently low leverage, high leverage inertia, and other leverage moments that are difficult to explain by taxes and distress costs will be inferred to have stronger agency conflicts.

4.5 Refinancing-cost benchmark

The model without agency conflicts estimates the refinancing-cost parameter using

$$\lambda_i = h(\alpha_\lambda + \varepsilon_i^\lambda). \quad (4)$$

Detailed interpretation: This benchmark is a diagnostic. If matching leverage dynamics requires refinancing costs far above observed issuance costs, the model is missing another force. The paper argues that agency conflicts supply this missing force.

4.6 Governance regressions

After estimating firm-level private benefits of control, the authors regress the predicted agency-cost estimates on governance variables:

$$\hat{\phi}_i = a + b' Governance_i + c' Controls_i + \epsilon_i. \quad (5)$$

Detailed interpretation: These regressions are not the source of the structural estimates. They are a validation exercise: the model-implied agency costs should be higher when observed governance is weaker.

5 Identification and empirical design

5.1 What identifies agency costs in the structural model

Agency costs are identified by the joint behavior of leverage moments. The paper does not infer agency costs from a single leverage level. It uses the conditional mean, median, volatility, persistence, skewness, kurtosis, range, changes in leverage, issuance frequencies, and default frequencies implied by the model.

Detailed interpretation: This is important because many parameters affect leverage levels. The distribution and dynamics of leverage provide additional identifying variation that helps separate refinancing costs from agency conflicts.

5.2 Why refinancing costs and agency costs are separable

Refinancing costs and agency costs affect leverage differently. Refinancing costs mostly create inactivity regions and persistence. Agency costs lower the manager's preferred target leverage and make underleverage persistent even when issuance costs are modest.

Detailed interpretation: The sensitivity columns in Table II show that the moments react differently to λ and ϕ . This is the central identification logic.

5.3 Why governance regressions matter

If the structural agency-cost estimates were only statistical artifacts, they would not systematically relate to independent governance variables. The negative relation with institutional ownership and board independence, and the positive relation with CEO tenure and antitakeover provisions, makes the interpretation more credible.

Detailed interpretation: The governance tests are an external validity check for the agency channel.

5.4 Limitations of the design

- The model is highly structured and relies on assumptions about cash-flow dynamics, taxes, bargaining, default, refinancing, and the functional form for heterogeneity.
- Governance regressions are correlational; they validate the direction of the estimates but do not randomly assign governance.
- The sample consists of public nonfinancial, nonregulated firms from 1992 to 2004, so the estimates may not generalize to private firms or other institutional environments.
- Capital structure may also reflect omitted factors such as product-market competition, debt covenants, managerial risk aversion, or strategic considerations not fully captured in the model.

Detailed interpretation: The paper's strength is that it asks a quantitative structural question and uses many moments. Its weakness is that the answer depends on the validity of the maintained structural model.

6 Section-by-section study guide

- **Introduction.** Sets up the low-leverage puzzle and motivates manager–shareholder conflicts as a quantitative explanation.
- **Section I: The Model.** Builds the dynamic tradeoff model with managerial private benefits, taxes, default, refinancing, liquidation, and renegotiation.
- **Section II: Model Calibration and Comparative Statics.** Shows how refinancing costs and agency costs affect target leverage, default leverage, recovery rates, and credit spreads.
- **Section III: Empirical Methodology.** Defines the sample, variables, structural parameters, simulated likelihood, and model moments.
- **Section IV: Estimation Results.** Estimates refinancing costs in the no-agency model and agency costs in the full model.
- **Section V: Governance and Agency Costs.** Relates structural agency-cost estimates to corporate governance proxies.
- **Conclusion.** Argues that modest agency conflicts have first-order effects on leverage levels and dynamics.

7 Tables and figures

7.1 Table I: Comparative Statics + Table II: Calibrated Leverage Distribution and Model Identification

Read:

- Table I reports how financing and default policies respond to changes in refinancing costs, private benefits, managerial ownership, bargaining power, liquidation costs, renegotiation costs, growth, volatility, and taxes.
- Higher private benefits of control lower target leverage sharply: when ϕ increases from 0.75% to 1.25%, target leverage falls from 37.61% to 24.12%.
- Higher managerial ownership raises target leverage because managers internalize more shareholder value.
- Table II compares empirical leverage moments to simulated moments and reports sensitivity to refinancing costs λ and private benefits ϕ .
- The baseline without agency conflicts produces average leverage that is too high. Raising refinancing costs or adding agency conflicts brings simulated leverage closer to the data.

Detailed interpretation: Table I gives the economic intuition. Private benefits reduce the manager’s demand for debt because debt disciplines free cash flow. Managerial ownership works in the opposite direction by aligning incentives. Table II shows that the identification is not based only on average leverage: refinancing costs and agency conflicts move dispersion, tails, autocorrelation, and event frequencies in different ways.

Economic message: Agency conflicts are not a cosmetic addition to the tradeoff model. They are a quantitatively meaningful force that lowers target leverage and helps match the full leverage distribution.

Table I
Comparative Statics

The table reports the main comparative statics of the dynamic model regarding the firm's financing and default policies, the recovery rate in default, and corporate credit yield spreads. We calibrate the base parametrization to the values observed in the data at the beginning of the sample period. Input parameter values for our base case are set as follows: risk-free interest rate $r = 4.21\%$, initial value of the cash flow shock $x_0 = 1$ (normalized), growth rate and volatility of the cash flow shock $m = 8.24\%$, $\mu = 0.67\%$, and $\sigma = 28.86\%$, corporate tax rate $\tau^c = 35\%$, tax rate on dividends $\tau^d = 11.6\%$, tax rate on interest income $\tau^i = 29.3\%$, liquidation costs $\alpha = 48.52\%$, renegotiation costs $\kappa = 0\%$, shareholder bargaining power $\eta = 50\%$, refinancing costs $\lambda = 1\%$, managerial ownership $\varphi = 7.47\%$, and private benefits of control $\phi = 1\%$.

	Quasi-Market Leverage (%) at			Target Credit Spread (%)	Recovery Rate (%)
	Restructuring	Target	Default		
Base	11.74	32.06	86.81	2.13	43.36
Cost of Debt Issuance (Base: $\lambda = 0.01$)					
$\lambda = 0.0075$	13.12	32.04	86.69	2.20	43.60
$\lambda = 0.0125$	10.58	31.91	86.93	2.05	43.11
Private Benefits of Control (Base: $\phi = 0.01$)					
$\phi = 0.0075$	15.79	37.61	85.92	2.75	46.01
$\phi = 0.0125$	7.17	24.12	87.91	1.46	40.32
Managerial Ownership (Base: $\varphi = 0.0747$)					
$\varphi = 0.05$	2.06	11.15	89.37	0.67	36.34
$\varphi = 0.1$	16.18	38.09	85.85	2.81	46.23
Shareholder Bargaining Power (Base: $\eta = 0.5$)					
$\eta = 0.25$	13.57	37.00	93.83	2.14	42.90
$\eta = 0.75$	9.91	27.09	78.72	2.11	43.82
Liquidation Costs (Base: $\alpha = 0.4852$)					
$\alpha = 0.45$	12.01	32.78	87.89	2.13	43.29
$\alpha = 0.55$	11.25	30.73	84.77	2.12	43.48
Renegotiation Costs (Base: $\kappa = 0$)					
$\kappa = 0.05$	9.70	27.35	88.91	1.76	39.04
$\kappa = 0.1$	8.36	24.05	90.66	1.53	35.58
Cash Flow Growth Rate (Base: $\mu = 0.0067$)					
$\mu = 0$	11.78	32.07	87.08	2.29	42.81
$\mu = 0.0125$	11.70	32.02	86.57	1.99	43.68
Cash Flow Volatility (Base: $\sigma = 0.2886$)					
$\sigma = 0.25$	13.30	34.16	86.04	1.61	46.38
$\sigma = 0.35$	9.96	29.47	87.79	3.09	39.52
Corporate Tax Rate (Base: $\tau_c = 0.35$)					
$\tau_c = 0.3$	0.28	3.73	89.50	0.28	36.78
$\tau_c = 0.4$	19.01	39.31	85.84	3.19	43.72

Table I: Comparative Statics

Table II
Calibrated Leverage Distribution and Model Identification

The table presents various data moments and a comparison with the corresponding moments simulated from the model. The table also reports sensitivities of the moments with respect to the model parameters. We obtain the model-implied moments and sensitivities by Monte Carlo simulation. The column titled "Empirical Moment" reports the value of the statistical moment in the data. The columns titled "Model Moments" report the values of the simulated statistical moments. We report model-implied moments for three different parametrizations. The parameters of interest are set to the following values in the baseline: $\lambda = 1\%$ and $\phi = 0\%$. The remaining parameters are calibrated as in Table I. The columns titled "Sensitivity" report the sensitivity of the model moments, $(\partial m / \partial \theta) / m$, at the baseline parameters for each of the structural parameters.

	Empirical Moment	Model Moments			Sensitivity	
		Baseline	λ higher	ϕ higher	λ (Baseline)	ϕ (Baseline)
		($\lambda = 1\%$, $\phi = 0\%$)	($\lambda = 8\%$, $\phi = 0\%$)	($\lambda = 1\%$, $\phi = 1.1\%$)		
Leverage:						
Mean	0.325	0.486	0.320	0.325	-3.574	-14.574
Median	0.292	0.463	0.280	0.279	-2.016	-17.512
SD	0.203	0.145	0.198	0.172	7.953	14.631
Skew	0.626	0.488	0.672	1.073	-22.990	29.074
Kurtosis	2.711	2.392	2.572	3.544	-4.903	6.324
IQR	0.297	0.213	0.303	0.218	11.393	15.151
Min	0.007	0.244	0.049	0.100	-20.272	-35.481
Max	0.980	0.842	0.861	0.872	0.314	1.971
Range	0.973	0.598	0.812	0.773	8.747	17.288
Autocorrelation 1 quarter	0.970	0.900	0.966	0.960	2.007	5.422
Autocorrelation 1 year	0.906	0.666	0.871	0.850	8.725	22.432
Changes in Leverage:						
Mean	0.003	0.002	0.001	0.001	-30.782	-25.843
Median	0.000	0.000	-0.001	-0.001	326.169	305.444
SD	0.057	0.060	0.049	0.044	1.859	-12.019
Skew	0.369	0.302	2.427	0.502	116.173	54.084
Kurtosis	11.165	3.533	21.659	4.980	29.899	14.976
IQR	0.051	0.074	0.042	0.048	-6.082	-16.245
Min	-0.634	-0.308	-0.256	-0.218	-11.797	-83.960
Max	0.569	0.267	0.420	0.241	-30.916	-38.661
Range	1.203	0.574	0.676	0.459	-21.753	-61.408
Autocorrelation 1 quarter	0.076	-0.061	-0.013	-0.023	-18.471	-70.106
Autocorrelation 1 year	0.008	-0.030	-0.012	-0.020	-59.120	-2.320
Event Frequencies:						
Pr(default)	-	0.589	0.212	0.159	-25.465	-51.733
Pr(issuance)	-	2.476	0.527	1.151	-49.890	-36.730
Issue size (%)	-	0.150	0.381	0.156	50.482	9.344

Table II: Calibrated Leverage Distribution and Model Identification

7.2 Figure 1: Monte Carlo Simulation Precision and Accuracy

Read:

- Panel A plots the simulated log-likelihood as the number of Monte Carlo draws increases.
- Panel B shows simulation imprecision falling sharply as the number of draws rises.
- Panel C shows simulation bias falling with more simulation draws.
- The figure justifies the simulation design used to estimate the structural model.

Detailed interpretation: The maximum likelihood estimator relies on simulated moments or simulated likelihood components. If simulation noise were large, the structural estimates could be unstable. The figure shows that precision and bias improve as the simulation size increases, supporting the reliability of the numerical procedure.

Economic message: The paper's empirical claims require not only a good economic model but also a reliable computational implementation. Figure 1 is a numerical credibility check.

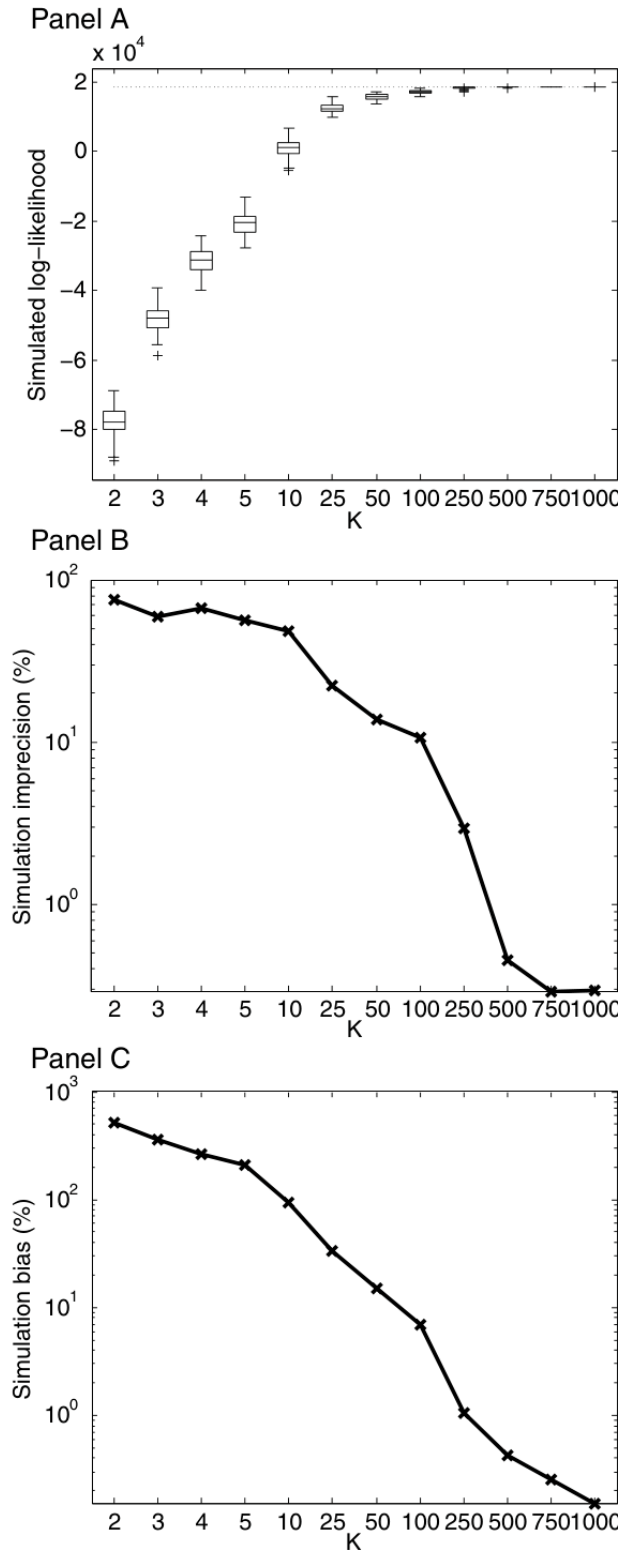


Figure 1: Monte Carlo Simulation Precision and Accuracy

7.3 Table III: Variable Definitions + Table IV: Descriptive Statistics

Read:

- Table III defines the accounting, market, compensation, ownership, governance, and macro variables used in the empirical work.
- Leverage is measured as book debt over assets minus book equity plus market equity, with an alternate book-debt definition for robustness.
- Managerial incentives combine direct managerial ownership and option-based pay-performance sensitivity.
- Table IV reports descriptive statistics for 13,159 firm-quarter observations.
- Mean market leverage is 0.32, median leverage is 0.29, and the standard deviation is 0.20, highlighting large leverage heterogeneity.
- The average institutional ownership is 0.60, board independence is 0.61, and CEO tenure is 8.67 years.

Detailed interpretation: These tables show that the empirical exercise is not purely structural abstraction. It maps model primitives to observable firm characteristics and governance measures. The descriptive statistics also show why a model must explain both leverage levels and large cross-sectional variation.

Economic message: The data contain substantial variation in leverage and governance, making it possible to connect structural agency costs to observed governance mechanisms.

Table III
Variable Definitions

Variable (Data Source)	Definition
Financial Indicators (Compustat):	
Book debt	Liabilities total (item 181) + Preferred stock (item 10) – Deferred taxes (item 35)
Book debt (alternate)	Long-term debt (item 9) + Debt in current liabilities (item 34)
Book equity	Assets total (item 9) – Book debt
Book equity (alternate)	Assets total (item 6) – Book debt (alternate)
Leverage	Book debt/(Assets total (item 6) – Book equity + Market value (item 25 * item 6))
Leverage (alternate)	Book debt (alternate)/(Assets total (item 6) – Book equity (alternate) + Market value (item 25 * item 6))
Return on assets	(EBIT (item 18) + Depreciation (item 14))/Assets total (item 6)
Market-to-book	(Market value (item 25 * item 6) + Book debt)/Assets total (item 6)
Tangibility	Property, plant, and equipment total net (item 8)/Assets total (item 6)
Size	log(Sales net (item 12))
R&D	Research and development expenses (item 46)/Assets total (item 6)
Earnings Growth (I/B/E/S):	
EBIT growth rate	Mean analyst forecast for long-term growth rate per SIC-2 industry
Volatility and Beta (CRSP):	
Equity volatility	Standard deviation of monthly equity returns over past 5 years
Market model beta	Market model beta based on monthly equity returns over past 5 years
Executive Compensation (ExecuComp):	
Managerial incentives	See Appendix C
Managerial ownership	Shares owned/Shares outstanding for the five highest paid executives
Managerial delta	See Appendix C
CEO tenure	Current year – year became CEO
EBIT growth rate (alternate)	Five-year least squares annual growth rate of operating income before depreciation
Blockholders (IRRC Blockholders):	
Blockholder ownership	Fraction of stock owned by outside blockholders
Directors (IRRC Directors):	
Board independence	Number of independent directors/total number of directors
Board committees	Sum of four dummy variables for existence of independent (more than 50% of committee directors are independent) audit, compensation, nominating, and corporate governance committee
Antitakeover Provisions (IRRC Governance):	
E-index	Six antitakeover provisions index by Bebchuk, Cohen, and Ferrell (2009)
G-index	24 antitakeover provisions index by Gompers, Ishii, and Metrick (2003)
Institutional Ownership (Thompson Financial):	
Institutional Ownership	Fraction of stock owned by institutional investors
Economy Indicators (FED):	
Term premium	Difference between 10-year and 1-year Treasury bond yield
Default premium	Difference between corporate yield spread (all industries) of Moody's BAA and AAA ratings

Table IV
Descriptive Statistics

The table presents descriptive statistics for the main variables used in the estimation. The sample is based on Compustat quarterly Industrial files, ExecuComp, CRSP, I/B/E/S, IRRC governance, IRRC blockholders, IRRC directors, and Thompson Financial. Table III provides a detailed definition of the variables.

	Mean	SD	25%	50%	75%	Observations
Leverage (y)	0.32	0.20	0.16	0.29	0.46	13,159
Leverage (alternate)	0.20	0.19	0.04	0.16	0.31	13,159
EBIT growth rate ($\tilde{m}$)	0.20	0.06	0.15	0.19	0.23	13,159
EBIT volatility (σ)	0.29	0.13	0.19	0.26	0.35	13,159
CAPM beta (β)	1.06	0.51	0.70	1.01	1.34	13,159
Liquidation costs (α)	0.51	0.12	0.45	0.50	0.58	13,159
Financial Characteristics:						
Return on assets	4.47	2.41	2.93	4.19	5.69	13,159
Market-to-book	2.05	1.27	1.23	1.64	2.39	13,159
Tangibility	0.34	0.22	0.16	0.28	0.47	13,159
Firm size	5.58	1.20	4.74	5.50	6.35	13,159
R&D	0.22	0.80	0.00	0.00	0.00	13,159
Ownership Structure:						
Institutional ownership	0.60	0.17	0.49	0.62	0.73	11,727
Blockholder ownership	0.09	0.13	0.00	0.00	0.16	13,159
Managerial Characteristics:						
Managerial incentives (φ)	0.07	0.09	0.02	0.04	0.08	13,159
Managerial ownership (φ^B)	0.05	0.08	0.00	0.01	0.05	13,159
Managerial delta	7.13	13.00	1.11	2.88	7.20	10,895
CEO tenure	8.67	8.78	2.42	5.92	11.90	13,159
Antitakeover Provisions:						
E-index	2.35	1.35	1.00	2.00	3.00	10,828
G-index	9.31	2.77	7.00	9.00	11.00	10,853
Board Structure:						
Board independence	0.61	0.18	0.50	0.63	0.75	8,665
Board committees	2.49	1.10	2.00	2.00	3.00	6,504
Macro Indicators:						
Term premium	0.01	0.01	0.00	0.01	0.02	13,159
Default premium	0.01	0.00	0.01	0.01	0.01	13,159

Table III: Variable Definitions

Table IV: Descriptive Statistics

7.4 Table V: Refinancing Cost Estimates Without Agency Conflicts + Table VI: Structural Parameter Estimates With Agency Conflicts

Read:

- Table V estimates a model without agency conflicts. The implied mean refinancing cost is 12.63% of total debt outstanding, with a median of 9.44%.
- Because the model-implied restructuring threshold multiplier is about two, this corresponds to roughly 25% of issue size, which is implausibly high relative to observed issuance costs.
- Table VI estimates the model with agency conflicts. The mean agency cost is 1.55% of equity value and the median is 0.45%.
- The distribution of agency costs is right-skewed: many firms have small estimated agency conflicts, but a nontrivial subset has large conflicts.

Detailed interpretation: Table V is a falsification exercise for a pure adjustment-cost explanation. It shows that the no-agency model can force-fit leverage dynamics only by assigning unrealistically large issuance costs. Table VI shows that introducing plausible managerial private benefits allows the model to explain the same patterns with much smaller frictions.

Economic message: Small agency costs can substitute for implausibly large refinancing costs in explaining conservative debt policy and leverage inertia.

Table V
Refinancing Cost Estimates in the Model without Agency Conflicts

The table presents the structural parameters characterizing the cost of refinancing, λ , in a dynamic capital structure model without agency conflicts ($\phi_i = 0$), defined as

$$\lambda_i = h(\alpha_\lambda + \epsilon_i^\lambda),$$

where $h = \Phi \in [0, 1]$ is the standard normal cumulative distribution function and $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$, $i = 1, \dots, N$. Panel A reports the parameter estimates. Cluster-robust t -statistics that adjust for cross-sectional correlation in each time period and industry-clustered t -statistics are reported in parentheses. Panel B reports distributional characteristics of the predicted model-implied cost of refinancing, $\hat{\lambda}_i = \hat{E}(\lambda_i | y_i; \theta)$. The refinancing cost estimates across firms are expressed in percent. The number of observations is 13,159.

Panel A: Parameter Estimates									
	α_λ			σ_λ			$\ln \mathcal{L}$		
Coefficient	-1.594			0.903			2,403		
t -stat (time clustered)	(-56.90)			(44.58)					
t -stat (industry clustered)	(-46.23)			(33.85)					
Panel B: Refinancing Cost Estimates Across Firms (%)									
	Mean	SD	Skewness	Kurtosis	5%	25%	50%	75%	95%
$\hat{\lambda}_i$	12.63	12.22	1.74	6.14	1.10	3.71	9.44	16.66	42.19

reports the point estimates. Cluster-robust t -statistics that adjust for cross-

Table V: Refinancing Cost Estimates in the Model without Agency Conflicts

Table VI
Structural Parameter Estimates in the Model with Agency Conflicts

The table presents estimation results of the structural parameters characterizing agency conflicts, defined by

$$\phi_i = h(\alpha_\phi + \epsilon_i^\phi),$$

where $h = \Phi \in [0, 1]$ is the standard normal cumulative distribution function and $\epsilon_i^\phi \sim \mathcal{N}(0, \sigma_\phi^2)$, $i = 1, \dots, N$. Panel A reports the parameter estimates. Cluster-robust t -statistics that adjust for cross-sectional correlation in each time period and industry-clustered t -statistics are reported in parentheses. The number of observations is 13,159. Panel B reports distributional characteristics of the predicted private benefits of control $\hat{\phi}_i = \hat{E}(\phi_i | y_i; \theta)$. In brackets we report these agency costs expressed as a fraction of equity value. The private benefits and cost estimates across firms are expressed in percent.

Panel A: Parameter Estimates									
	α_ϕ				σ_ϕ			$\ln \mathcal{L}$	
Coefficient	−2.595				0.791			7,436	
t -stat (time clustered)	(−23.05)				(18.85)				
t -stat (industry clustered)	(−21.05)				(12.17)				
Panel B: Agency Cost Estimates Across Firms (%)									
	Mean	SD	Skewness	Kurtosis	5%	25%	50%	75%	95%
$\widehat{\phi}_i$	1.55	3.04	4.03	22.35	0.02	0.13	0.45	1.71	7.93
	[1.36]	[2.85]	[4.46]	[27.34]	[0.02]	[0.10]	[0.37]	[1.36]	[6.39]

Table VI: Structural Parameter Estimates in the Model with Agency Conflicts

7.5 Figure 2: Predicted Agency Costs Across Firms + Figure 3: Cost of Debt and Underleverage

Read:

- Figure 2 plots the distribution of estimated private benefits of control across firm-quarters.
- The distribution is concentrated near zero but has a long right tail, indicating that agency conflicts are limited for many firms but severe for some.
- Figure 3 decomposes the marginal benefit and marginal cost of debt from shareholder and managerial perspectives.

- Shareholders choose leverage where shareholder marginal benefits and costs intersect, l_S^* .
- Managers choose lower leverage, l_M^* , because the disciplining effect of debt enters their objective as a cost.
- The shaded area between the manager's and shareholder's optimal leverage represents the cost of underleverage.

Detailed interpretation: Figure 2 is the empirical distribution of the model's core agency-cost parameter. Figure 3 gives the intuition behind the parameter's effect: debt may maximize shareholder value but reduce managerial private benefits. This difference creates a wedge between shareholder-optimal and manager-optimal leverage.

Economic message: The low-leverage puzzle is a governance wedge: managers do not perceive debt's disciplining effect as a benefit in the same way shareholders do.

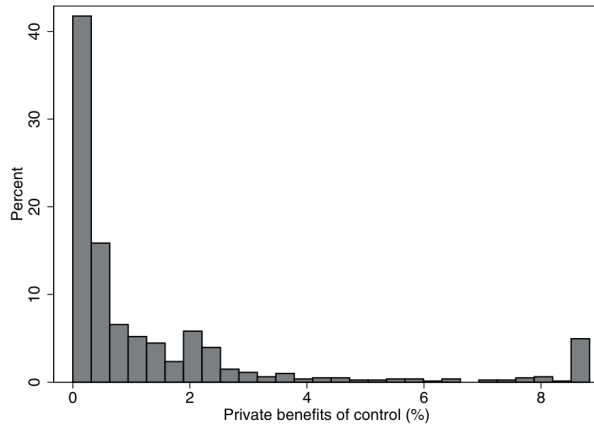


Figure 2. Predicted agency costs across firms. The figure shows the predicted private benefit of control $\mathbb{E}(\phi_i | y_i, x_i; \theta)$, for firms $i = 1, \dots, N$, in the dynamic capital structure model.

Figure 2: Predicted Agency Costs Across Firms

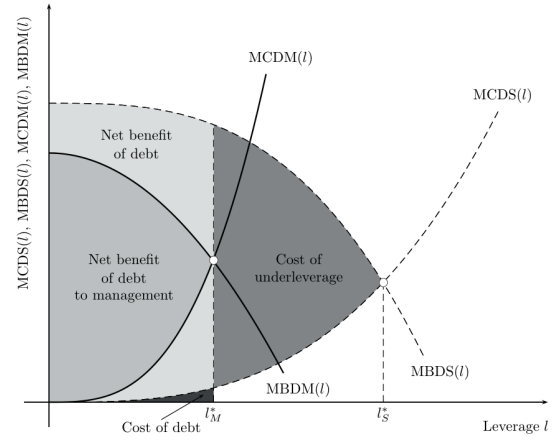


Figure 3. Cost of debt and underleverage. The figure plots the marginal benefit of debt to management, $MBDM(l)$, and the marginal benefit of debt to shareholders, $MBDS(l)$, and the marginal cost of debt, $MCDM(l)$ and $MCDS(l)$, as a function of leverage l .

Figure 3: Cost of Debt and Underleverage

7.6 Table VII: Distribution of Debt Costs Across Firms + Table VIII: Robustness of Agency Cost Estimates

Read:

- Table VII decomposes debt benefits and costs from the perspectives of shareholders and managers.
- Gross benefit of debt to shareholders averages 10.69% of shareholder wealth, while cost of debt to shareholders averages 1.07%.
- Cost of debt to managers averages 3.38% of managerial utility, about three times the shareholder cost.
- More than half of the managerial cost comes from the disciplining effect of debt.
- Table VIII shows that the agency-cost estimates are robust to alternative restructuring costs, ownership measures, renegotiation costs, leverage definitions, time-varying parameters, and link functions.

Detailed interpretation: Table VII translates the model into value units. The central point is that managers and shareholders do not face the same marginal cost schedule for debt. Table VIII addresses a concern that the agency-cost estimate may be an artifact of one narrow specification. The estimates vary but remain economically meaningful across alternatives.

Economic message: The agency channel is robust and economically interpretable: debt creates value for shareholders partly because it imposes discipline that managers dislike.

Table VII
Distribution of Debt Costs Across Firms

The table reports the distributional characteristics of the net benefit of debt to managers (NBDM) and shareholders (NBDS), the gross benefit of debt to managers (GBDM) and shareholders (GBDS), the cost of debt to managers (CDM) and shareholders (CDS), as well as the cost of the disciplining effect of debt to the manager (DE). All values referring to shareholders are reported as a percentage of shareholder wealth. All values referring to managers are reported as a percentage of managerial utility.

	Mean	SD	Skewness	Kurtosis	5%	25%	50%	75%	95%
NBDM	6.87	8.47	0.31	10.28	0.00	0.00	4.17	11.20	24.60
NBDS	9.62	9.91	0.76	2.57	0.00	0.01	9.13	14.73	28.35
GBDM	10.25	10.63	0.76	2.53	0.00	0.01	9.80	15.86	30.41
GBDS	10.69	11.03	0.75	2.53	0.01	0.01	10.15	16.46	31.65
CDM	3.38	4.58	4.33	59.21	0.01	0.03	2.16	5.01	12.30
CDS	1.07	1.15	0.80	2.63	0.00	0.00	0.88	1.71	3.32
DE	1.82	3.08	2.56	10.50	0.01	0.02	0.24	2.61	8.18

Table VII: Distribution of Debt Costs Across Firms

Table VIII
Robustness: Agency Cost Estimates under Alternative Specification

The table summarizes distributional characteristics of the fitted private benefits of control, $\phi_i = \mathbb{E}(\phi_i | y_i; \theta)$, $i = 1, \dots, N$, in a number of alternative specifications. In Appendix A, we derive explicit expressions for $\hat{\phi}_i$. In brackets we report private benefits of control expressed as a fraction of equity value. The cost estimates across firms are expressed in percent.

	Mean	SD	Skewness	Kurtosis	5%	25%	50%	75%	95%
Restructuring Cost $\lambda = 0.50\%$									
$\hat{\phi}_i$	1.82 [1.64]	3.51 [3.43]	3.93 [4.34]	21.41 [25.81]	0.02 [0.01]	0.12 [0.09]	0.52 [0.42]	1.99 [1.58]	8.54 [7.56]
Alternate Ownership Measure φ^E									
$\hat{\phi}_i$	1.08 [0.97]	2.68 [2.55]	4.71 [4.90]	29.28 [31.64]	0.01 [0.00]	0.04 [0.03]	0.18 [0.15]	0.88 [0.66]	6.20 [5.05]
Renegotiation Cost $\kappa = 15\%$									
$\hat{\phi}_i$	1.10 [0.97]	2.29 [2.15]	4.10 [4.22]	22.96 [23.47]	0.01 [0.01]	0.09 [0.07]	0.31 [0.25]	1.04 [0.86]	6.10 [5.28]
Alternate Definition of Leverage									
$\hat{\phi}_i$	3.10 [3.92]	3.98 [5.62]	1.61 [1.73]	5.96 [5.96]	0.03 [0.02]	0.23 [0.19]	0.96 [0.82]	7.08 [7.15]	9.81 [13.76]
Parameters θ^* Set to Time-Varying Estimates θ^*_t									
$\hat{\phi}_i$	1.67 [1.50]	3.42 [3.10]	4.07 [3.87]	22.98 [20.81]	0.01 [0.01]	0.09 [0.07]	0.40 [0.34]	1.77 [1.44]	8.20 [7.74]
Logit Specification for Link Function h									
$\hat{\phi}_i$	1.77 [1.57]	3.80 [3.57]	4.67 [4.91]	29.21 [32.23]	0.02 [0.02]	0.13 [0.10]	0.43 [0.37]	1.80 [1.51]	8.21 [7.30]

Table VIII: Robustness: Agency Cost Estimates under Alternative Specification

7.7 Table IX: Conditional Moment Tests + Table X: Simulation Evidence on Leverage Inertia and Mean Reversion

Read:

- Table IX compares empirical leverage moments to simulated moments and reports conditional moment tests.
- The joint Wald test does not reject the model, although some individual moments differ.
- The model matches key persistence moments well: quarterly autocorrelation is 0.905 in the model versus 0.916 in the data; annual autocorrelation is 0.680 versus 0.737.
- Table X shows that simulated leverage inertia resembles the empirical inertia documented by Welch and Strebulaeu.
- The model also implies partial mean reversion of leverage, with a two-stage estimated speed around 11% per year.

Detailed interpretation: Table IX is the formal goodness-of-fit test: the model is not only estimating parameters but also being asked to match moments that matter in the capital-structure literature. Table X shows that the model reproduces the slow adjustment patterns that are central to empirical leverage dynamics.

Economic message: The model performs well precisely where earlier dynamic capital-structure puzzles are most salient: leverage is persistent, adjustment is partial, and firms do not mechanically rebalance to target leverage.

Table IX
Conditional Moment Tests for Goodness-of-Fit

The table reports the results from conditional moment (CM) tests that use conditional moment restrictions for testing goodness-of-fit. The CM tests check whether the difference between the real data moments listed in each row and the simulated data moments based on our SML estimates is equal to zero. The test statistic for each individual moment is reported in the third column next to the corresponding moment, and the associated p -value is in the last column. The Wald test statistic tests for joint fit of the model.

	Empirical Moment	Simulated Model Moment	CM Test Statistic	p -value
Leverage:				
Mean	0.322	0.325	(-0.26)	[0.79]
Median	0.317	0.309	(1.38)	[0.17]
SD	0.068	0.101	(-19.19)	[0.00]
Skew	0.220	0.755	(-23.78)	[0.00]
Kurtosis	2.308	3.596	(-32.31)	[0.00]
Range	0.216	0.400	(-34.00)	[0.00]
IQR	0.100	0.136	(-12.23)	[0.00]
Min	0.224	0.157	(11.24)	[0.00]
Max	0.440	0.557	(-14.26)	[0.00]
Autocorrelation 1 quarter	0.916	0.905	(0.23)	[0.82]
Autocorrelation 1 year	0.737	0.680	(1.21)	[0.23]
Wald test of joint hypothesis H_0 : all moments equal			(0.34)	[1.00]

Table IX: Conditional Moment Tests for Goodness-of-Fit

Table X
Simulation Evidence: Leverage Inertia and Mean Reversion

The table provides simulation evidence on leverage inertia and mean reversion. Panel A reports parameter estimates from Fama-MacBeth (1973) regressions on leverage in levels, similar to Welch (2004). The basic specification is as follows:

$$L_t = \alpha_0 + \alpha_1 L_{t-k} + \alpha_2 IDR_{t-k,t} + \epsilon_t,$$

where L is the leverage ratio, IDR is the implied debt ratio defined in Welch (2004), and k is the time horizon. Coefficients reported are means over 1,000 simulated data sets. Below our estimated coefficients we report the coefficients on IDR in Welch (2004) and Strebulaev (2007). Panel B reports parameter estimates from Fama-MacBeth (1973) regressions on leverage changes, similar to Fama and French (2002). The basic specification is as follows:

$$L_t - L_{t-1} = \alpha + \lambda_1 T L_{t-1} + \lambda_2 L_{t-1} + \epsilon_t,$$

where L is, again, the leverage ratio and TL is the target leverage ratio. In the first specification, TL is determined in a prior stage by running a cross-sectional regression of leverage on various determinants. In the second specification, TL is set to the model-implied target leverage ratio. Coefficients are means over 1,000 simulated data sets.

Panel A: Leverage Inertia				
	Lag k in Years			
	1	3	5	10
Coefficient Estimates in Simulated Data				
Implied debt ratio, $IDR_{t-k,t}$	1.02	0.87	0.75	0.56
Leverage, L_{t-k}	-0.06	0.05	0.14	0.30
Constant	0.01	0.02	0.03	0.04
R^2	0.96	0.89	0.83	0.74
$IDR_{t-k,t}$ Coefficients in the Literature				

Table X: Simulation Evidence: Leverage Inertia and Mean Reversion

7.8 Table XI: The Determinants of Agency Conflicts

Read:

- Table XI regresses predicted private benefits of control on governance and firm characteristics.
- Institutional ownership is strongly negatively related to agency conflicts.
- Independent blockholder ownership is generally negatively related to agency conflicts.
- CEO tenure is positively related to agency conflicts, consistent with entrenchment.
- Dictatorship indicators based on antitakeover provisions are positively related to agency conflicts.
- Board independence is negatively related to agency conflicts, consistent with monitoring by independent directors.
- Size is negatively related to agency conflicts, while market-to-book is positively related.

Detailed interpretation: This table is the main external validation of the structural agency-cost estimates. It shows that estimated agency costs behave like governance-based theories predict: stronger monitoring reduces them, entrenchment increases them, and the cross-sectional explanatory power is substantial.

Economic message: The structural parameter $\hat{\phi}$ is not just a statistical residual. It is economically related to governance institutions that plausibly affect managerial discretion.

Table XI
The Determinants of Agency Conflicts

The table summarizes the determinants of agency conflicts across firms. The dependent variable is the predicted value of private benefits of control, $\hat{\phi}_i = \hat{\mathbb{E}}(\phi_i | y_i; \theta)$ for $i = 1, \dots, N$, where θ are the parameters estimated in Section IV.B. and $\hat{\phi}_i$ are expressed in basis points. In columns (1) to (4), we report estimation results from cross-sectional regressions. Specifications (1) and (4) use the entire sample. Missing values are imputed with zero and dummy variables that take a value of one for missing values are included in the regression. Specifications (2) and (3) only use observations with no missing data items. In specification (3) we drop the variables with the most missing values from the regression. All specifications are estimated including industry fixed effects. Standard errors are adjusted for sampling error in the generated regressands (see Newey and McFadden (1994, p. 2183)) and are reported in parentheses. Statistical significance at the 10%, 5%, and 1% level is indicated by *, **, and ***, respectively.

	(1)	(2)	(3)	(4)
Institutional ownership	-201.53*** (51.26)	-205.26*** (58.01)	-233.12*** (54.82)	-215.48*** (52.68)
Independent blockholder ownership	-112.70* (68.00)	-117.11 (72.89)	-113.84* (67.25)	-107.89 (67.75)
CEO tenure	4.48*** (1.26)	4.81*** (1.70)	5.62*** (1.51)	4.23*** (1.27)
E-index—Dictatorship	268.16** (114.82)	283.87* (148.98)	250.30* (146.14)	—
G-index—Dictatorship	—	—	—	68.22 (95.82)
Board independence	-181.16*** (56.29)	-182.44** (71.12)	—	-176.72*** (55.92)
Board committees	0.65 (9.25)	0.99 (9.74)	—	0.22 (9.46)
Managerial delta (Quartile 1)	-2.80 (17.42)	-13.17 (19.19)	-14.72 (19.00)	-5.84 (17.74)
Managerial delta (Quartile 4)	4.02*** (0.83)	4.48*** (0.96)	4.42*** (0.90)	4.02*** (0.84)
Returns on assets	6.91 (5.11)	3.09 (6.62)	5.05 (5.83)	8.43* (5.07)
Market-to-book	32.80*** (10.21)	34.49** (15.43)	30.75** (14.22)	24.85*** (9.33)
Asset tangibility	-72.71* (37.66)	-48.98 (44.65)	-46.38 (41.79)	-43.08 (44.57)
Size	-51.99*** (7.68)	-54.04*** (9.06)	-54.43*** (8.10)	-52.14*** (10.31)
R&D	-9.46 (10.04)	-11.52 (13.20)	-13.34 (12.85)	-7.54 (10.27)
Observations	809	569	634	809
R^2	0.35	0.37	0.35	0.35

Table XI: The Determinants of Agency Conflicts

8 Detailed result map

- **Comparative statics.** Higher private benefits reduce leverage; higher managerial ownership raises leverage; corporate taxes increase leverage; volatility raises spreads and lowers recovery.
- **Identification.** Leverage moments are sensitive to λ and ϕ in different ways, allowing the model to distinguish refinancing frictions from agency conflicts.
- **No-agency benchmark.** Without agency conflicts, the model requires implausibly high refinancing costs to fit leverage dynamics.
- **Agency model.** With agency conflicts, modest private benefits of control match leverage levels and dynamics.
- **Distribution of agency costs.** Agency costs are right-skewed; most firms have small conflicts, but a minority have large conflicts.
- **Governance relation.** Agency costs are lower with stronger monitoring and higher with entrenchment proxies.
- **Moment fit.** The model matches key persistence and inertia moments and passes a joint goodness-of-fit test.

9 Short summary

- The paper develops a dynamic tradeoff model with manager–shareholder conflicts.
- Managers own equity but also capture private benefits from free cash flow.
- Debt disciplines managers by reducing free cash flow and private benefits.
- Because managers dislike this disciplining effect, they choose less debt than shareholders would prefer.
- The model includes taxes, refinancing costs, liquidation costs, renegotiation, and default.
- The empirical sample contains 13,159 firm-quarter observations for 809 firms from 1992 to 2004.
- The model is estimated using many leverage moments, not just average leverage.
- A no-agency model needs implausibly high refinancing costs to match leverage dynamics.
- A model with agency conflicts estimates average agency costs of about 1.5% of equity value.
- These modest agency costs resolve the low-leverage puzzle and explain leverage inertia.
- The estimated agency costs are related to governance: stronger monitoring lowers them and entrenchment raises them.
- The paper provides a structural explanation for persistent firm-specific leverage heterogeneity.

10 Conclusion

10.1 Core conclusion

Morellec, Nikolov, and Schürhoff show that manager–shareholder conflicts have first-order effects on capital structure. A dynamic tradeoff model with refinancing costs alone can generate inertia, but it cannot quantitatively explain conservative leverage without implausibly high issuance costs. Adding modest managerial private benefits produces leverage targets and leverage dynamics that resemble the data.

10.2 Economic intuition

The core intuition is that debt has different meanings for shareholders and managers. For shareholders, debt creates tax shields and disciplines free cash flow. For managers, the disciplining effect is partly a cost because it reduces cash flows that can be diverted or consumed as private benefits. Thus, managers choose lower leverage than shareholders would choose. The amount of underleverage depends on the severity of agency conflicts and the strength of governance mechanisms.

10.3 Methodological contribution

The paper’s key methodological contribution is to infer agency costs from the entire dynamic behavior of leverage. This is stronger than showing that governance proxies correlate with leverage. The structural model must explain leverage levels, dispersion, autocorrelation, changes, tails, and adjustment behavior. The resulting agency-cost estimates are then validated using independent governance variables.

10.4 Relation to the capital-structure literature

The paper reconciles two strands of research. Dynamic tradeoff models explain inertia and adjustment bands, while agency models explain managerial reluctance to lever up. The paper shows that both forces are necessary: refinancing costs create slow adjustment around a target, and agency conflicts shift the target downward.

10.5 Corporate governance implications

The results imply that governance can affect corporate financing policy by changing the effective private benefits of control. Institutional investors, independent blockholders, and independent boards appear to reduce managerial discretion and therefore reduce the agency cost of underleverage. CEO tenure and antitakeover protections increase agency conflicts and contribute to lower leverage.

10.6 Limitations

The estimates depend on the structural model's assumptions about cash-flow dynamics, taxes, refinancing, bargaining, liquidation, renegotiation, and managerial private benefits. Governance regressions are validating correlations rather than randomized tests. The paper also abstracts from some potentially important factors such as managerial risk aversion, covenant design, debt maturity structure, and product-market strategy.

10.7 Takeaway

The paper's lasting message is that capital structure is partly a governance outcome. Firms do not merely balance taxes and bankruptcy costs; their leverage choices are made by managers whose private objectives differ from shareholder value. Even small private benefits of control can produce large and persistent deviations from shareholder-optimal leverage.